Statistics & Probability

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Measures of Central Tendency & Dispersion**

A machine's failure probability increases each hour it runs; X is the number of failures in a day. Is X binomially distributed? Show your complete algebraic work.

Final Answer: No — trials are not independent / probability is not constant. The conditions for a binomial distribution (fixed independent trials with constant probability) are not satisfied here.

2 **[Foundational]** • **Target: Probability Rules & Conditional Probability**

A discrete random variable X has probability distribution given by

$P(X=5)=k, P(X=4)=5k, P(X=3)=3k, P(X=2)=4k, P(X=1)=2k$, where k is a constant. Find k.

Final Answer: 1/15. Since probabilities sum to 1: $(1+5+3+4+2)k=1 \Rightarrow 15k=1 \Rightarrow k=1/15$

3 **[Foundational]** • **Target: Discrete Probability Distributions**

A discrete random variable X has probability distribution given by

$P(X=6)=4k, P(X=2)=3k, P(X=3)=5k, P(X=4)=2k, P(X=1)=3k$, where k is a constant. Find k.

Final Answer: 1/17. Since probabilities sum to 1: $(4+3+5+2+3)k=1 \Rightarrow 17k=1 \Rightarrow k=1/17$

4 **[Foundational]** • **Target: Binomial Distribution & Exam Applications**

A discrete random variable X has probability distribution given by

$P(X=2)=4k, P(X=6)=3k, P(X=1)=k, P(X=4)=4k, P(X=5)=4k$, where k is a constant. Find k.

Final Answer: 1/16. Since probabilities sum to 1: $(4+3+1+4+4)k=1 \Rightarrow 16k=1 \Rightarrow k=1/16$

5 **[Foundational]** • **Target: Measures of Central Tendency & Dispersion**

A discrete random variable X has probability distribution given by

$P(X=0)=5k, P(X=1)=k, P(X=6)=4k, P(X=3)=2k, P(X=5)=5k$, where k is a constant. Find k.

Final Answer: 1/17. Since probabilities sum to 1: $(5+1+4+2+5)k=1 \Rightarrow 17k=1 \Rightarrow k=1/17$

6 **[Foundational]** • **Target: Probability Rules & Conditional Probability**

A discrete random variable X has probability distribution given by

$P(X=6)=2k, P(X=2)=3k, P(X=0)=k, P(X=1)=3k, P(X=3)=4k$, where k is a constant. Find k.

Final Answer: 1/13. Since probabilities sum to 1: $(2+3+1+3+4)k=1 \Rightarrow 13k=1 \Rightarrow k=1/13$

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7 [Foundational] • Target: Discrete Probability Distributions

A discrete random variable X has probability distribution given by

$P(X=0)=2k, P(X=3)=k, P(X=1)=5k, P(X=6)=k, P(X=5)=5k$, where k is a constant. Find k.

Final Answer: 1/14. Since probabilities sum to 1: $(2+1+5+1+5)k=1 \Rightarrow 14k=1 \Rightarrow k=1/14$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Binomial Distribution & Exam Applications

A discrete random variable X has probability distribution given by

$P(X=0)=4k, P(X=3)=3k, P(X=2)=5k, P(X=6)=4k, P(X=1)=k$, where k is a constant. Find k.

Final Answer: 1/17. Since probabilities sum to 1: $(4+3+5+4+1)k=1 \Rightarrow 17k=1 \Rightarrow k=1/17$

9 [Intermediate] • Target: Measures of Central Tendency & Dispersion

A discrete random variable X has probability distribution given by

$P(X=2)=2k, P(X=3)=k, P(X=5)=2k, P(X=4)=4k, P(X=6)=4k$, where k is a constant. Find k.

Final Answer: 1/13. Since probabilities sum to 1: $(2+1+2+4+4)k=1 \Rightarrow 13k=1 \Rightarrow k=1/13$

10 [Intermediate] • Target: Probability Rules & Conditional Probability

A discrete random variable X has probability distribution given by

$P(X=1)=4k, P(X=2)=k, P(X=3)=4k, P(X=4)=k$, where k is a constant. Find k.

Final Answer: 1/10. Since probabilities sum to 1: $(4+1+4+1)k=1 \Rightarrow 10k=1 \Rightarrow k=1/10$

11 [Intermediate] • Target: Discrete Probability Distributions

A discrete random variable X has probability distribution given by

$P(X=3)=k, P(X=2)=3k, P(X=1)=k, P(X=4)=3k, P(X=6)=k$, where k is a constant. Find k.

Final Answer: 1/9. Since probabilities sum to 1: $(1+3+1+3+1)k=1 \Rightarrow 9k=1 \Rightarrow k=1/9$

12 [Intermediate] • Target: Binomial Distribution & Exam Applications

A discrete random variable X has probability distribution given by

$P(X=1)=k, P(X=2)=k, P(X=3)=k, P(X=4)=4k$, where k is a constant. Find k.

Final Answer: 1/7. Since probabilities sum to 1: $(1+1+1+4)k=1 \Rightarrow 7k=1 \Rightarrow k=1/7$

13 [Intermediate] • Target: Measures of Central Tendency & Dispersion

A discrete random variable X has probability distribution given by

$P(X=1)=k, P(X=2)=k, P(X=3)=2k$, where k is a constant. Find k.

Final Answer: 1/4. Since probabilities sum to 1: $(1+1+2)k=1 \Rightarrow 4k=1 \Rightarrow k=1/4$

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14 [Intermediate] • Target: Probability Rules & Conditional Probability

A discrete random variable X has probability distribution given by

$P(X=1)=3k, P(X=2)=3k, P(X=3)=k$, where k is a constant. Find k.

Final Answer: 1/7. Since probabilities sum to 1: $(3+3+1)k=1 \Rightarrow 7k=1 \Rightarrow k=1/7$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Discrete Probability Distributions • Cross-Lesson: 3.01 Graphs & Transformations

A discrete random variable X has probability distribution given by

$P(X=1)=k, P(X=2)=3k, P(X=3)=k$, where k is a constant. Find k.

Final Answer: 1/5. Since probabilities sum to 1: $(1+3+1)k=1 \Rightarrow 5k=1 \Rightarrow k=1/5$

16 [Challenging] • Target: Binomial Distribution & Exam Applications • Cross-Lesson: 4.01 Coordinate Geometry

A discrete random variable X has probability distribution given by

$P(X=1)=2k, P(X=2)=3k, P(X=3)=2k$, where k is a constant. Find k.

Final Answer: 1/7. Since probabilities sum to 1: $(2+3+2)k=1 \Rightarrow 7k=1 \Rightarrow k=1/7$

17 [Challenging] • Target: Measures of Central Tendency & Dispersion • Cross-Lesson: 5.01 Trigonometry

A discrete random variable X has probability distribution given by

$P(X=1)=2k, P(X=2)=k, P(X=3)=2k$, where k is a constant. Find k.

Final Answer: 1/5. Since probabilities sum to 1: $(2+1+2)k=1 \Rightarrow 5k=1 \Rightarrow k=1/5$

18 [Challenging] • Target: Probability Rules & Conditional Probability • Cross-Lesson: 6.01 Differentiation & Integration

A discrete random variable X has probability distribution given by

$P(X=1)=k, P(X=2)=2k, P(X=3)=3k$, where k is a constant. Find k.

Final Answer: 1/6. Since probabilities sum to 1: $(1+2+3)k=1 \Rightarrow 6k=1 \Rightarrow k=1/6$

19 [Challenging] • Target: Discrete Probability Distributions • Cross-Lesson: 8.01 Functions: Composite & Inverse

A discrete random variable X has probability distribution given by

$P(X=1)=3k, P(X=2)=2k, P(X=3)=2k$, where k is a constant. Find k.

Final Answer: 1/7. Since probabilities sum to 1: $(3+2+2)k=1 \Rightarrow 7k=1 \Rightarrow k=1/7$

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20 [Challenging] • Target: Binomial Distribution & Exam Applications • Cross-Lesson: 9.01 Circular Measure

A research team is studying the impact of different fertilizer types on crop yield. They conducted an experiment with two fertilizer types, A and B, and recorded the yield in kilograms per plot. The data is summarized in the two-way table below. The total number of plots is 200. | Fertilizer Type | Yield < 50 kg | Yield ≥ 50 kg | Total | |---|---|---|---| | A | 30 | 70 | 100 | | B | 50 | 50 | 100 | | Total | 80 | 120 | 200 | Given that a plot was selected at random and its yield was less than 50 kg, what is the probability that it received fertilizer type B? Provide your answer as a fraction in its simplest form.

Final Answer: 5/8. Let Y be the event that the yield is less than 50 kg, and B be the event that the fertilizer type is B. We are asked to find the conditional probability P(B|Y). From the table, we have: Number of plots with yield < 50 kg (event Y) = 80. Number of plots with yield < 50 kg AND fertilizer type B (event B ∩ Y) = 50. The formula for conditional probability is P(B|Y) = P(B ∩ Y) / P(Y). Alternatively, using the counts directly: P(B|Y) = (Number of plots with yield < 50 kg and fertilizer type B.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Measures of Central Tendency & Dispersion

The numbers $4, 7, 9, 12, 18$ have a mean of $\bar{x} = 10$. Find the standard deviation σ correct to 2 decimal places.

a. 4.73 ✓

b. 5.21

c. 4.12

d. 3.85

Squared deviations: $(4-10)^2 = 36, (7-10)^2 = 9, (9-10)^2 = 1, (12-10)^2 = 4, (18-10)^2 = 64$. Sum $= 114$. Variance $= 114/5 = 22.8$. $\sigma = \sqrt{22.8} \approx 4.73$.

22 [Foundational] • Target: Measures of Central Tendency & Dispersion

A data set of 20 observations has $\sum x = 240$ and $\sum x^2 = 3280$. Find the variance of the data set.

a. **20** ✓

b. 16

c. 24

d. 12

Mean $\bar{x} = 240/20 = 12$. Variance $= \frac{\sum x^2}{n} - \bar{x}^2 = \frac{3280}{20} - 12^2 = 164 - 144 = 20$.

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23 [Foundational] • Target: Measures of Central Tendency & Dispersion

In a box-and-whisker plot, the lower quartile is $Q_1 = 24$ and the upper quartile is $Q_3 = 42$. According to the $1.5 \times \text{IQR}$ rule, what is the boundary for outliers at the upper end?

a. 69 ✓

b. 60

c. 65

d. 72

$\text{IQR} = Q_3 - Q_1 = 42 - 24 = 18$. Upper boundary $= Q_3 + 1.5 \times \text{IQR} = 42 + 1.5(18) = 42 + 27 = 69$.

24 [Foundational] • Target: Probability Rules & Conditional Probability

Two independent events A and B have $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \cup B)$.

a. 0.70 ✓

b. 0.90

c. 0.20

d. 0.65

For independent events, $P(A \cap B) = P(A)P(B) = 0.4 \times 0.5 = 0.2$. Then

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.4 + 0.5 - 0.2 = 0.70$.

25 [Foundational] • Target: Probability Rules & Conditional Probability

Given $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$, find the conditional probability $P(A \mid B)$.

a. 0.60 ✓

b. 0.50

c. 0.30

d. 0.75

$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{0.3}{0.5} = 0.60$.

26 [Foundational] • Target: Probability Rules & Conditional Probability

A bag contains 4 red balls and 6 blue balls. Two balls are drawn at random without replacement. What is the probability that both balls are red?

a. $\frac{2}{15}$ ✓

b. $\frac{4}{25}$

c. $\frac{1}{6}$

d. $\frac{1}{5}$

$P(R_1) = 4/10 = 2/5$. $P(R_2 \mid R_1) = 3/9 = 1/3$. Combined probability $= \frac{2}{5} \times \frac{1}{3} = \frac{2}{15}$.

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27 [Foundational] • Target: Discrete Probability Distributions

A discrete random variable X has probability function $P(X = x) = kx$ for $x \in \{1, 2, 3, 4\}$. Find the value of k.

a. $\frac{1}{10}$ ✓

b. $\frac{1}{6}$

c. $\frac{1}{4}$

d. $\frac{1}{12}$

Sum of probabilities must equal 1: $k(1 + 2 + 3 + 4) = 1 \Rightarrow 10k = 1 \Rightarrow k = 1/10$.

28 [Foundational] • Target: Discrete Probability Distributions

Find the expectation $E(X)$ for the distribution where $P(X = 1) = 0.2, P(X = 2) = 0.5, P(X = 3) = 0.3$.

a. 2.1 ✓

b. 2.0

c. 2.3

d. 1.9

$E(X) = \sum xP(X = x) = 1(0.2) + 2(0.5) + 3(0.3) = 0.2 + 1.0 + 0.9 = 2.1$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Discrete Probability Distributions

For a discrete random variable X with $E(X) = 4$ and $E(X^2) = 25$, what is the variance $\text{Var}(X)$?

a. 9 ✓

b. 5

c. 16

d. 21

$\text{Var}(X) = E(X^2) - [E(X)]^2 = 25 - 4^2 = 25 - 16 = 9$.

30 [Intermediate] • Target: Binomial Distribution & Exam Applications

A random variable $X \sim B(8, 0.25)$. Find $P(X = 2)$ correct to 4 decimal places.

a. 0.3115 ✓

b. 0.2076

c. 0.2670

d. 0.1038

$P(X = 2) = \binom{8}{2}(0.25)^2(0.75)^6 = 28 \times 0.0625 \times 0.1779785 \approx 0.3115$.

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31 [Intermediate] • Target: Binomial Distribution & Exam Applications

If $X \sim B(100, 0.4)$, find the mean and variance of X.

a. **Mean** $= 40$**, Variance** $= 24$ ✓

b. Mean $= 40$, Variance $= 16$

c. Mean $= 50$, Variance $= 25$

d. Mean $= 40$, Variance $= 4.9$

Mean $\mu = np = 100(0.4) = 40$. Variance $\sigma^2 = np(1-p) = 100(0.4)(0.6) = 24$.

32 [Intermediate] • Target: Binomial Distribution & Exam Applications

A multiple-choice quiz has 5 questions, each with 4 options and exactly one correct answer. If a student guesses at random on all 5 questions, what is the probability of passing with at least 3 correct answers?

a. $\frac{53}{512} \approx 0.1035$ ✓

b. 0.2500

c. 0.0879

d. 0.1563

$P(X \geq 3) = P(3) + P(4) + P(5) = \binom{5}{3}(1/4)^3(3/4)^2 + \binom{5}{4}(1/4)^4(3/4) + \binom{5}{5}(1/4)^5 = (10 \times 9 + 5 \times 3 + 1)/1024 = (90 + 15 + 1)/1024 = 106/1024 = 53/512 \approx 0.1035$ ▪

33 [Intermediate] • Target: Measures of Central Tendency & Dispersion

The two-way table shows the results of a survey of 100 students on their preferred mode of transport to school and their year group. | | Car | Bus | Walk | Total | |--------------|-----|-----|------|-------| | Year 1 | 15 | 20 | 10 | 45 | | Year 2 | 25 | 18 | 12 | 55 | | Total | 40 | 38 | 22 | 100 | What is the probability that a randomly selected student is a Year 1 student who prefers to travel by car?

a. **0.15** ✓

b. 0.33

c. 0.38

d. 0.45

The total number of students surveyed is 100. The number of Year 1 students who prefer to travel by car is 15. Therefore, the probability of selecting a Year 1 student who prefers to travel by car is the number of Year 1 students who prefer to travel by car divided by the total number of students. P(Year 1 and Car) = Number of (Year 1 and Car) / Total Number of Students = 15 / 100 = 0.15.

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34 [Intermediate] • Target: Probability Rules & Conditional Probability

A bag contains 18 counters, of which 9 are red. A counter is drawn at random. Find the probability that it is red.

a. 1/4

b. **1/2 ✓**

c. 15/17

d. 7/10

Applying Statistics & Probability principles, $P(\text{red}) = \frac{9}{18} = 1/2$ Therefore, the verified correct option is 1/2.

35 [Intermediate] • Target: Discrete Probability Distributions

A discrete random variable X has distribution $P(X = 1) = 2/5, P(X = 2) = 1/5, P(X = 3) = 2/5$. Given $Y = 5X - 3$, find $E(Y)$.

a. 6

b. **7 ✓**

c. 8

d. 13

Applying Discrete Random Variables principles, $E(X) = 2$. $E(Y) = aE(X) + b = 5(2) - 3 = 7$ Therefore, the verified correct option is 7.

36 [Intermediate] • Target: Binomial Distribution & Exam Applications

A discrete random variable X has distribution $P(X = 1) = 3/10, P(X = 2) = 2/5, P(X = 3) = 3/10$. Given $Y = 4X - 4$, find $E(Y)$.

a. 3

b. 5

c. **4 ✓**

d. 12

Applying Discrete Random Variables principles, $E(X) = 2$. $E(Y) = aE(X) + b = 4(2) - 4 = 4$ Therefore, the verified correct option is 4.

37 [Intermediate] • Target: Measures of Central Tendency & Dispersion

Given $X \sim B(58, 9/20)$, find $E(X)$.

a. 133/5

b. **261/10 ✓**

c. 251/10

d. 271/10

Applying Binomial Distribution principles, $E(X) = np = 58 \times 9/20 = 261/10$ Therefore, the verified correct option is 261/10.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Probability Rules & Conditional Probability • Cross-Lesson: 6.01 Differentiation & Integration
Given $X \sim B(46, 9/20)$, find $E(X)$.

a. **207/10 ✓**
b. 106/5
c. 197/10
d. 217/10

Applying Binomial Distribution principles, $E(X) = np = 46 \times 9/20 = 207/10$ Therefore, the verified correct option is 207/10.

39 [Challenging] • Target: Discrete Probability Distributions • Cross-Lesson: 8.01 Functions: Composite & Inverse
Given $X \sim B(37, 7/20)$, find $E(X)$.

a. **259/20 ✓**
b. 239/20
c. 269/20
d. 279/20

Applying Binomial Distribution principles, $E(X) = np = 37 \times 7/20 = 259/20$ Therefore, the verified correct option is 259/20.

40 [Challenging] • Target: Binomial Distribution & Exam Applications • Cross-Lesson: 9.01 Circular Measure
Given $X \sim B(47, 3/20)$, find $E(X)$.

a. **141/20 ✓**
b. 121/20
c. 161/20
d. 151/20

Applying Binomial Distribution principles, $E(X) = np = 47 \times 3/20 = 141/20$ Therefore, the verified correct option is 141/20.

41 [Challenging] • Target: Measures of Central Tendency & Dispersion • Cross-Lesson: 1.01 Indices, Surds & Quadratics
Given $X \sim B(75, 11/20)$, find $E(X)$.

a. 167/4
b. 169/4
c. 161/4
d. **165/4 ✓**

Applying Binomial Distribution principles, $E(X) = np = 75 \times 11/20 = 165/4$ Therefore, the verified correct option is 165/4.

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42 [Challenging] • Target: Probability Rules & Conditional Probability • Cross-Lesson: 10.01 Series & Binomial Theorem

Given $X \sim B(75, 13/20)$, find $E(X)$.

a. 191/4

b. **195/4 ✓**

c. 199/4

d. 197/4

Applying Binomial Distribution principles, $E(X) = np = 75 \times 13/20 = 195/4$ Therefore, the verified correct option is 195/4.

43 [Challenging] • Target: Discrete Probability Distributions • Cross-Lesson: 11.01 Kinematics

Given $X \sim B(17, 1/10)$, find $E(X)$.

a. 27/10

b. **17/10 ✓**

c. 11/5

d. 7/10

Applying Binomial Distribution principles, $E(X) = np = 17 \times 1/10 = 17/10$ Therefore, the verified correct option is 17/10.

44 [Challenging] • Target: Binomial Distribution & Exam Applications • Cross-Lesson: 12.01 Forces & Equilibrium

Given $X \sim B(26, 3/20)$, find $E(X)$.

a. 49/10

b. 22/5

c. 29/10

d. **39/10 ✓**

Applying Binomial Distribution principles, $E(X) = np = 26 \times 3/20 = 39/10$ Therefore, the verified correct option is 39/10.

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45 [Challenging] • Target: Measures of Central Tendency & Dispersion • Cross-Lesson: 13.01 Newton's Laws

Students are surveyed until the first one who owns a car is found; X is the number of students asked. Is X binomially distributed?

a. Yes — but only if $n > 30$

b. Yes — fixed, independent trials with constant success probability

c. No — the number of trials is not fixed

d. **No — trials are not independent / probability is not constant ✓**

The conditions for a binomial distribution (fixed independent trials with constant probability) are not satisfied here.

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